

# KALVIN CHANG

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## EDUCATION

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### University of California, Berkeley

Berkeley, CA

Doctor of Philosophy in Computer Science

Aug 2025 - May 2030 (expected)

Advisor: Alane Suhr, GPA: 4.0/4.0

*Selected courses: Data-Centric LLMs, Deep RL, Large-Scale Vision-Language & Action Models*

### Carnegie Mellon University (CMU)

Pittsburgh, PA

Master of Science in Language Technologies, School of Computer Science

Jan 2022 - May 2023

GPA: 4.28/4.33 (Rank: 1/18), Advisor: David Mortensen

*Selected courses: Speech Processing, Multilingual NLP, Computational Ethics of NLP,*

*ML for Structured Data (probabilistic graphical models), Phonetics & Phonemics, Phonology*

Bachelor of Science in Computer Science

Aug 2018 - Dec 2021

GPA: 3.67/4.0 (University Honors), Concentration in Human-Computer Interaction

*Courses: Design & Analysis of Algorithms, Distributed Systems, Designing Human-Centered Software*

## RESEARCH EXPERIENCE

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### Meta Superintelligence Labs, Voice Modeling Team

Menlo Park, CA

*Research Scientist Intern, Supervisors: Dr. Aparna Aketi, Yashesh Gaur*

Jun 2026 - present

- Post-training duplex conversational speech language model with RLHF

### Berkeley AI Research Lab

Berkeley, CA

*Graduate Research Assistant, Advisor: Prof. Alane Suhr*

Aug 2025 - present

- Pre-training textless speech language model with flow matching and post-training with RLHF

### Tencent AI Lab, Multimodal Understanding Team

Bellevue, WA

*AGI Research Intern, Supervisors: Dr. Dong Yu, Dr. Yiwon Shao*

Mar 2025 - May 2025

- Built a speech encoder with cross-lingual alignment without paired translation data (Chang *et al.* 2026)
- Trained a state-of-the-art Zipformer Chinese dialect ASR model with distributed training (32 GPUs)

### CMU WAVLab & ChangeLingLab

Pittsburgh, PA

*Visiting Scholar, Advisors: Prof. Shinji Watanabe, Prof. David Mortensen*

Aug 2024 - Feb 2025

- Secured a NSF grant (\$1,000,000) on in-context learning for low-resource ASR using historical linguistics
- Trained open source, speech-to-text foundation model for universal phone recognition (Li *et al.* 2025)
- Created 3 linguistic datasets for speech LM instruction-following benchmark (ICLR 2025)

### CMU Language Technologies Institute

Pittsburgh, PA

*Graduate Research Assistant, Advisor: Prof. David Mortensen*

Jan 2022 - May 2023

- Discovered bias in self-supervised speech models against African American English (Interspeech 2024)
- Collected Taiwanese corpus for end-to-end ASR with self-supervised speech features (ASRU 2023)
- Set state-of-the-art on protoform reconstruction with a Transformer encoder-decoder model (ACL 2023)
- Built a neural pipeline matching 87.6% of a linguist's language family tree (LChange 2023)
- Expanded open-source G2P package's coverage of 7 low-resource Chinese varieties, enabling the collection of 67,000+ pronunciations to address gap in comparative Chinese datasets (Coling 2022)
- Proposed metric learning & masked LM for learning phonetic word embeddings (LREC-Coling 2024)

## PUBLICATIONS

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- [**ACL 2026**] C. Li\*, **K. Chang\***, S. Bharadwaj, E. Yeo, K. Choi, J. Zhu, D. Mortensen, S. Watanabe. *POWSM: A Phonetic Open Whisper-Style Speech Foundation Model*. **Oral Presentation and Top 5%**.
- [**ACL Findings 2026**] R. Shim, K. Choi, **K. Chang**, M.H. Hsu, *et al.* *Linear Script Representations in Speech Foundation Models Enable Zero-Shot Transliteration*.
- [**ACL 2026**] S. Bharadwaj *et al.* *PRiSM: Benchmarking Phone Realization in Speech Models*.
- [**Interspeech 2025**] B. Yan *et al.* *CS-FLEURS: A Massively Multilingual and Code-Switched Speech Dataset*.
- [**ACL 2025**] A. Naik *et al.* *Programming by Example meets Historical Linguistics: A Large Language Model Based Approach to Sound Law Induction*
- [**ICLR 2025**] C. Huang *et al.* *Dynamic-SUPERB Phase-2: A Collaboratively Expanding Benchmark for Measuring the Capabilities of Spoken Language Models with 180 Tasks*.
- [**NAACL 2025**] K. Choi, E. Yeo, **K. Chang**, S. Watanabe, and D. Mortensen. *Leveraging Allophony in Self-Supervised Speech Models for Atypical Pronunciation Assessment*.
- [**Interspeech 2024**] **K. Chang\***, Y. Chou\*, J. Shi, H. Chen, N. Holliday, O. Scharenborg, and D. Mortensen. *Self-supervised Speech Representations Still Struggle with African American Vernacular English*. **Honorable Mention, Special Session on Responsible Speech Foundation Models**.
- [**IEEE ASRU 2023**] Y. Chou\*, **K. Chang\***, *et al.* *Evaluating Self-Supervised Speech Models on a Taiwanese Hokkien Corpus*.
- [**ACL 2023**] Y.M. Kim\*, **K. Chang\***, C. Cui, and D. Mortensen. *Transformed Protoform Reconstruction*. **Oral Presentation**.
- [**LChange 2023**] **K. Chang\***, N. Robinson\*, A. Cai\*, T. Chen, A. Zhang, and D. Mortensen. *Automating Sound Change Prediction for Phylogenetic Inference: A Tukanooan Case Study*. **Oral Presentation**.
- [**LREC-Coling 2024**] V. Zouhar\*, **K. Chang\***, C. Cui, N. Carlson, N. Robinson, M. Sachan, and D. Mortensen. *PWESuite: Phonetic Word Embeddings and Tasks They Facilitate*.
- [**LREC-Coling 2024**] R. Shim\*, **K. Chang\***, and D. Mortensen. *Phonotactic Complexity across Dialects*.
- [**Coling 2022**] **K. Chang**, C. Cui, Y.M. Kim, and D. Mortensen. *WikiHan: A New Comparative Dataset for Chinese Languages*.
- [preprint] **K. Chang**, Y. Shao, J. Li, D. Yu. *Towards Comprehensive Semantic Speech Embeddings for Chinese Dialects*.

\* denotes equal contribution.

## Academic Service

Organizer: UC Berkeley EECS Equal Access to Application Assistance Program

Reviewer: *Speech Communication*, NAACL 2025, ARR Oct. 2025, ACL 2026, ICASSP 2026

## WORK EXPERIENCE

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### Amazon Web Services

Seattle, WA

*Software Development Engineer, Test Generation Team*

*Aug 2023 - Aug 2024*

- Doubled # executable tests with runtime feedback in an LLM-based test generation platform
- Designed automatic evaluation suite for LLM-generated tests for 384 AWS services in an ECS container

*Software Development Engineer Intern, EC2 Quality Team*

*Jun 2021 - Aug 2021*

- Implemented pagination & presigned URL for website diagnosing EC2 Instances during on-call

### Cardinal Blue Software, Inc.

Taipei, Taiwan

*Server Developer Intern*

*Feb 2020 - Jul 2020*

- Transitioned PostgreSQL database to transactional pooling, increasing scalability 18-fold

## AWARDS

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|-------------------------------------------------------------------------------------------|------------------------------------------|
| UC Berkeley EECS Department Fellowship                                                    | <i>Aug 2025</i>                          |
| Apple AI Residency, Finalist (80 / 6000)                                                  | <i>Feb 2025</i>                          |
| Gates Cambridge Scholarship, University of Cambridge (35 / 600 U.S. recipients; declined) | <i>Jan 2025</i>                          |
| Honorable Mention, Responsible Speech Foundation Models, Interspeech                      | <i>Sep 2024</i>                          |
| University Honors, Carnegie Mellon University                                             | <i>May 2022</i>                          |
| Dean's List with High Honors, CMU School of Computer Science                              | <i>Fall 2020, Spring 2021, Fall 2021</i> |

## TEACHING EXPERIENCE

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**Teaching Assistant, Introduction to Machine Learning** *Fall 2022*

- Developed coursework for 400+ students, which covered Hidden Markov Models and gradient descent

**Teaching Assistant, Principles of Functional Programming** *Fall 2019, Fall 2020*

- Held office hours and labs for 200-student course covering structural induction and higher order functions

## ACTIVITIES

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**Researcher, SDAIA Winter School** *Dec 2024*

- Selected to attend the inaugural winter school on multimodal LLMs, designed as an extension to JSALT
- Decreased WER on ASR for codeswitching using Whisper prompted with in-context learning

**Leader, Computational Historical Linguistics Subgroup** *Jan 2023 - May 2023*

- Recruited and led team of 6 towards publication on phylogenetic inference (Chang *et al.* 2023)
- Mentored 3 freshmen who were new to machine learning, deep learning, NLP, and research

**The Impact Fellowship, Impact Labs** *Jan 2019*

- Selective (< 5%) two-week program to train software engineers working in tech for social good
- Connected with speakers and leaders from NGOs, social startups, and think tanks

**Mentor, Glen A. Wilson High School CODE Team** *2018 - present*

- Taught human-centered design, web dev (HTTP, API design) and data structures (big O, linked lists)
- Awarded one of two Leadership and Service Awards by Principal Dr. Danielle Kenfield
- Hosted Shark Tank to critique Congressional App Challenge ideas, emphasizing human-centered design
- Provided free college apps critiques, helping 2 high school students secure Amazon internships
- Encouraged students to find their interdisciplinary niche within CS (e.g. computational linguistics)
- Mentored twenty-five alumni across various universities (e.g. UC Berkeley, UCSD)

## TALKS

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|------------------------------------------------------------------------------|--------------------------|
| How AI Works, Wilson Hacks, Glen A. Wilson CODE                              | <i>March 21, 2026</i>    |
| Self-supervised Speech Repr. Still Struggle with AAVE, LMU NLP for Dialects  | <i>December 11, 2025</i> |
| AI: A Gentle Introduction, Wilson Hacks, Glen A. Wilson CODE                 | <i>March 15, 2025</i>    |
| Neural Reconstruction of Middle Chinese, CMU Language Technologies Institute | <i>May 3, 2023</i>       |
| Computer Science After High School, Wilson Hacks, Glen A. Wilson CODE        | <i>Apr 1, 2023</i>       |

## SKILLS

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|                               |                                                                                                        |
|-------------------------------|--------------------------------------------------------------------------------------------------------|
| <b>Software Engineering</b>   | Python, Java, Ruby, Go, C, AWS, HTML/CSS, JS, Git                                                      |
| <b>Machine Learning Tools</b> | PyTorch, vLLM, ESPnet, fairseq, transformers, SLURM                                                    |
| <b>Languages</b>              | English (native), Mandarin Chinese (native),<br>Spanish (California Bilingual Seal), Taiwanese Hokkien |